

Practice Data Warehousing Assignment

Version: 1.0 – 09/09/2026

A. General Information and Submission

- This assignment requires the Oracle database.
- A marking rubric has been provided for the practice assignment

B. Problem Description

MonCity is one of the largest smart cities in the world, and was developed by Monash University for education and research purposes. In MonCity, there are four different zone areas (ZoneA, ZoneB, ZoneC, ZoneD). A wide range of self-driving cars is provided for Monash students and staff members to travel between different zone areas or within the same zone area.

MonCity has an existing operational database that maintains and stores all of the self-driving car-related information, such as the booking, maintenance, and accident records required for management's daily operation. However, in order to improve work efficiency, management at MonCity has decided to hire your team of data warehouse engineers to design and develop a data warehouse that can quickly generate reports based on their needs. Management at MonCity wants to generate reports to keep track of the **bookings, accidents and maintenance information**, such as calculating statistics of booking records and accident records, which can be used for self-driving car analyses later. For the accident details, they are particularly interested in the **number of accidents per self-driving car** (e.g., the number of accidents caused by Car01).

MonCity's operational database tables can be found in the [MonCity](#) account. You can, for example, execute the following query:

```
Select * From MonCity.<table_name>;
```

The data definition of each table in MonCity is as follows:

Table Name	Attributes and Data Types		Notes
FACULTY	FACULTYID	VARCHAR	This table stores faculty information. One faculty can have multiple passengers.
	FACULTYNAME	VARCHAR	

	ZONE	VARCHAR	
RESEARCHCENTER	CENTERID	VARCHAR	This table stores information about research centres. One research centre may have several maintenance teams.
	CENTERNAME	VARCHAR	
	PHONE	NUMBER	
	OPENINGHOUR	VARCHAR	
PASSENGER	PASSENGERID	VARCHAR	This table stores passenger information. Each passenger is from a certain faculty and can have one or multiple booking records.
	PASSENGERNAME	VARCHAR	
	PASSENGERROLE	VARCHAR	
	PASSENGERGENDER	VARCHAR	
	PASSENGERAGE	NUMBER	
	FACULTYID	VARCHAR	
ERROR	ERRORCODE	VARCHAR	This table stores accident error information. One error code may involve more than one accident.
	ERRORMESSAGE	VARCHAR	
MAINTENANCETYPE	MAINTENANCETYPE	VARCHAR	This table stores maintenance type information. One maintenance type may involve more than one maintenance record.
	MAINTENANCEDESCRIPTION	VARCHAR	
BOOKING	BOOKINGID	VARCHAR	This table stores the information of booking records. One booking record belongs to a certain passenger and car.
	REGISTRATIONNO	VARCHAR	
	BOOKINGDATE	DATE	
	DEPARTUREZONE	VARCHAR	
	DESTINATIONZONE	VARCHAR	

	PASSENGERID	VARCHAR	
CAR	REGISTRATIONNO	VARCHAR	This table stores the information of self-driving cars. One car may have more than one booking, one accident, and one maintenance record.
	CARMODEL	VARCHAR	
	MANUFACTURINGYEAR	NUMBER	
	CARBODYTYPE	VARCHAR	
	NUMSEATS	NUMBER	
MAINTENANCE	MAINTENANCEID	VARCHAR	This table stores each car's maintenance history. Each maintenance record belongs to a certain car, maintenance type and maintenance team.
	REGISTRATIONNO	VARCHAR	
	MAINTENANCEDATE	DATE	
	MAINTENANCETYPE	VARCHAR	
	MAINTENANCECOST	NUMBER	
	TEAMID	VARCHAR	
ACCIDENTINFO	ACCIDENTID	VARCHAR	This table stores the information of each accident, including accident ID, accident zone, etc. Each accident can be categorised into a certain error code.
	ACCIDENTZONE	VARCHAR	
	CAR_DAMAGE_SEVERITY	VARCHAR	
	ERRORCODE	VARCHAR	
CARACCIDENT	REGISTRATIONNO	VARCHAR	This table stores information about accidents and all related cars. One accident can involve more than one car.
	ACCIDENTID	VARCHAR	
MAINTENANCETEAM	TEAMID	VARCHAR	This table stores information about maintenance teams. One maintenance team may have more than one maintenance record. Each maintenance team may
	TEAMLEADER	VARCHAR	
	PHONE	NUMBER	

			be from more than one research center.
BELONGTO	TEAMID	VARCHAR	This table stores the relationship between research centres and maintenance teams.
	CENTERID	VARCHAR	

C. Task

The assignment is divided into **FIVE** main tasks:

C1.Design a data warehouse for the above MonCity database.

You are required to create a data warehouse for the **MonCity** database. The management is interested in the following fact measures:

- Number of booking records
- Total maintenance cost
- Number of maintenance records
- Number of accident records

The following are some possible dimension attributes that you may need in your data warehouse:

- Booking time period: by month
- Faculty
- Passenger age group: Young adults (18-35 years old); middle-aged adults (36-59 years old); old-aged adults (60 years old and older)
- Car body type (including number of seats)
- Research centre
- Maintenance type
- Accident Zone (based on accident zone in the ACCIDENTINFO table)
- Error code
- Car damage severity

For each attribute, you may apply your own design decisions on specifying a range or group where this is not already detailed above, but make sure to specify them in your submission.

C2.Preparation stage.

Before you start designing the data warehouse, you have to ensure that you have explored the operational database and have done sufficient data cleaning. Once you have done the data cleaning process, you are required to explain what strategies you have taken to explore and clean the data

The outputs of this task for **Report** are:

- a) If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning).

C3.Drawing a star/snowflake schema.

The star schema for this data warehouse may contain multi-fact(s). You need to identify the fact measures, dimensions, and attributes of the star/snowflake schema. The following queries might help you to identify the fact measures and dimensions:

- How many booking records were made by passengers from the IT faculty in July by using bus?
- How many booking records were made by each passenger age group?
- How many maintenance records were listed that include at least one maintenance team from research centre 4 (CE04)?
- List the total maintenance cost of each maintenance type for mini bus.
- How many accidents were recorded in the ZoneA for Car01?
- List the number of accidents for each error code at ZoneB for Car06.
- List the number of accidents with severe damage inflicted on Car05.

You should pay attention to the granularity of your fact tables. You are required to create **Version-1** of star/snowflake schemas at the **highest level of aggregation**. For **Version-2**, you are required to suggest how the fact table granularity could be changed to achieve **Level 0 (no aggregation)**. You are not required to create a schema diagram for Version-2.

Version Name	Level
Version-1	High aggregation
Version-2	No aggregation (Level 0)

The star/snowflake schema of both versions you created might contain **bridge tables**, **determinant dimensions**, and/or **temporal dimensions**. If you are using bridge tables, make sure to include the weight factor and list aggregate attributes. If you are

using determinant dimensions, make sure the correct notation has been used (broken lines for the table boundary) and you must provide the reasons for your choice(s). You can use different temporal data warehousing techniques for the temporal dimensions. If there are any, you must provide the reasons for your choice(s).

The outputs of task C3 for **Report** are:

- b) A star/snowflake schema diagram for design Version-1 (You can use [Lucidchart](#) or [Draw.io](#) to draw the star schema.)
- c) List suggestion of increasing the granularity to Level 0 of your fact table for design Version-2. You only need to identify which attribute(s) should be changed or added (a schema diagram and implementation for Version-2 is not required.)
- d) An explanation of the difference among SCD types 0, 1, 2, 3, 4, and 6. Provide an example of a temporal dimension and describe which SCD type(s) would be appropriate for that dimension in Design Version-1 or Design Version-2. If your design includes a temporal dimension, you may explain that instead.

Note:

The star/snowflake schema design, including the fact measures and dimensions, is entirely based on the specific questions we provide for each exercise and assignment.

It is essential to closely follow the question requirements and design the star schema accordingly. The fact measures and dimensions should align precisely with the data and analysis needed to address the given problem.

Please ensure that you thoroughly understand the question prompts and the expected outcomes before proceeding with the star schema design. **Any deviation from the provided question may lead to inaccurate results and impact your assessment.**

Asking the tutor to check whether your design is correct or not will not be answered, as it is part of the assessment in this assignment.

C4.Implement the **Version-1** of the star/snowflake schemas using SQL.

You are required to implement the Version-1 of star/snowflake schema that you have drawn in Task C3. This implies that you need to create the fact and dimension tables in SQL. The output is a series of SQL statements to perform this task. You will also need to show that this task has been carried out successfully.

Note:

- If your account is full, you will need to drop all of the tables that you have previously created during labs.

- If you have dropped all tables in your account and you still encounter the ORA-01536: space quota exceeded for tablespace 'TABLE_NAME', please check your SQL code to ensure you have properly joined all tables. This issue is mainly caused when you do not join the tables properly, as the number of records is multiplied during the process.

The outputs of this task for **Report** are:

- e) Screenshots of the table structure you created for Design Version-1, including the dimension tables and fact tables.

A sample of screenshots of the table structure

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ADDRESS_ID	NUMBER	Yes	(null)	1	(null)
2	STREET_NUMBER	NUMBER	Yes	(null)	2	(null)
3	STREET_NAME	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)
4	SUBURB	VARCHAR2(20 BYTE)	Yes	(null)	4	(null)
5	STATE	VARCHAR2(10 BYTE)	Yes	(null)	5	(null)
6	POSTCODE	NUMBER	Yes	(null)	6	(null)

C5. Data Analysis

Due to the successful operation of the MonCity project, Monash University decided to invest further in the self-driving car project. As a data warehouse engineer:

- You are required to provide **a suggestion with valid supporting data** to help management decide on how to improve current self-driving car projects, such as which car body type should be further invested in based on the Booking, maintenance, and accident records, or which error code has contributed to the most number of accidents, etc.;

Note: The above examples are for guidance only. You may identify and analyse other relevant areas based on the MonCity data.

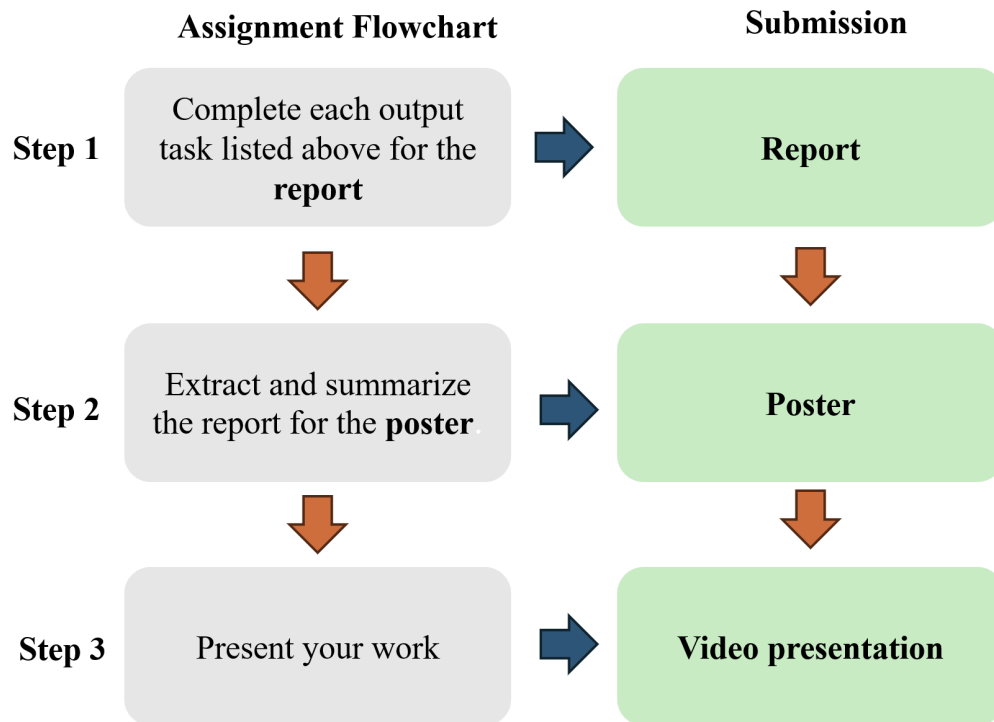
- If you have a different opinion on the self-driving car project, provide a reason why the project is not worth further investment.

Note: The above suggestion must contain **valid supporting data from the MonCity database and data warehouse**. This could be tables, graphs, or charts. Please use a maximum of 500 words for your suggestions.

The outputs of this task for **Report** are:

1. **Findings report:** Provide a detailed explanation of your descriptive analysis findings, including any significant observations, trends, or patterns identified in the data.
2. **Power BI Dashboard:** Create a dashboard using Power BI Desktop/Web that includes visualisations relevant to your data analysis. Export the dashboard to PDF and merge it with your report

D.Submission Checklist



Step 1: Report (35% of the total score)

A combined pdf file save as: YourstudentID_A2_report.pdf, containing all of the above tasks:

- A. Cover page
- B. If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning).
- C. A star/snowflake schema diagram for Version-1
- D. List suggestions to increase the granularity of your fact tables for design Version-2
- E. An explanation of the difference among SCD types 0, 1, 2, 3, 4, and 6. Provide an example of a temporal dimension and describe which SCD type(s) would be appropriate for that dimension in Design Version-1 or Design Version-2.
- F. Screenshots of the table structure you created for Design Version-1 only, including the dimension table and fact tables.
 - a. **SQL file for creating the star/snowflake schema is NOT required in submission**
- G. Findings report: A detailed explanation of your findings, including any significant observations or patterns identified during the analysis.
- H. One dashboard created in Power BI, exported as a PDF and included in the report

Step 2: Poster (35% of the total score)

**One page standard A4 poster in PDF format to save as:
YourstudentID_A2_poster.pdf**

Extract key information from the report you created and present it in a one-page poster. The poster must be in standard A4 size and in PDF format, which can be either landscape or portrait. The content should be clear and easy to understand. Avoid using technical jargon or complex language. Review the poster before submission to ensure it effectively communicates the key messages of your report.

Note:

Ensure the poster content is consistent with the key structure and findings of your report, and choose an appropriate layout that effectively organizes the information in a clear and logical manner. Maintain a good balance of text and visuals to enhance readability, and ensure all visuals are relevant and support the content of the poster. Label all visuals clearly and provide captions where necessary. Avoid overcrowding the poster with too much text or too many visuals, and ensure the poster is free of any grammatical or typographical errors.

Key guidance of design a poster:

- What is the main theme/objective of the poster that you want to express?
- Who is your target audience for this poster?
- Do you really need all the details from your report on this poster?

Step 3: Video presentation (30% of the total score)

**A five minute video presentation in mp4 format save as:
YourstudentID_A2_video.mp4**

Based on the report and poster you have created, present your design and findings in a five-minute video presentation. Ensure you thoroughly understand both the report and the poster to effectively extract and communicate the key points.